

JaeSeong Kim

UNDERGRADUATE RESEARCHER · LLM INTERNALS · REPRESENTATION ANALYSIS · EFFICIENT LLM

Semyung University, South Korea

✉ mmmqp1010@gmail.com | 🏠 Website 🤗 HuggingFace (1, 2)

Summary

LLMs internalize knowledge and reasoning within billions of parameters, yet how this capability is organized internally remains largely unexplored. What geometric structures do representations form? How do layers coordinate? Why do different models converge to similar structures on the same task? I seek to answer these questions and translate that understanding into practical value. By dynamically composing the internal representations of different models, we can construct new capabilities; by tracking representational changes in real time, we can detect jailbreak attacks before they succeed. Ultimately, I believe that sufficient data and probabilistic modeling can give rise to an intrinsic understanding of the world. If the internal representations of LLMs already reflect the structure of the world, then understanding and extending them is the most realistic path toward AGI.

Research Interests

- LLM Internals: representation space analysis, latent geometry, inter-layer coordination, task-selective convergence
- Efficient LLM: training-free adaptation, calibration, pruning, distillation, selective fine-tuning
- Safety & Alignment: jailbreak detection, multilingual euphemistic coating, module attack, unlearning

Education

Semyung University

B.SC. IN COMPUTER SCIENCE

- IT Engineering College, Dept. of Computer Science
- Data Intelligence Lab (Advisor: Prof. Suan Lee)
- GPA: 4.12 / 4.50

Jecheon, South Korea

Mar. 2023 – Expected Feb. 2027

Research Experience

Data Intelligence Lab, Semyung University

UNDERGRADUATE RESEARCHER (1ST AUTHOR × 22)

Advisor: Prof. Suan Lee

Mar. 2023 – Present

- **4 first-author papers accepted at top-tier venues in 2026:** UAI, KDD (Datasets & Benchmarks Track), ECML PKDD, ACL Rolling Review (Findings)
- **20 accepted publications with 4 papers under review**
- LLM Internals: representation space analysis, latent geometry, inter-layer coordination tracking, task-selective convergence
- Efficient LLM: pruning, distillation, selective fine-tuning, training-free adaptation and calibration
- Safety & Alignment: jailbreak detection, multilingual euphemistic coating, module attack, unlearning
- Reasoning & Benchmarking: small-scale reasoning models (KO-SmallThinker), K-GovExam benchmark

Decision Making Lab, Korea University

RESEARCH INTERN

Advisor: Prof. Byung-Jun Lee

Apr. 2026 – Present

Publications

20 accepted · 4 under review · 3 paper awards · 4 at top-tier venues

SELECTED PUBLICATIONS — TOP-TIER VENUES

1. **JaeSeong Kim** and Suan Lee, “Same Benchmark, Same Subspace: Task-Selective Convergence in LLM Representations,” **UAI 2026** (Conf. on Uncertainty in Artificial Intelligence).
2. **JaeSeong Kim**, ChaeHwan Lim, SangHyeon Gil, and Suan Lee, “EuraGovExam: A Multilingual Multimodal Benchmark from Real-World Civil Service Exams,” **KDD 2026** (ACM SIGKDD), Datasets & Benchmarks Track. [arXiv]
3. **JaeSeong Kim** and Suan Lee, “The Latent Hivemind: Do LLMs Converge in How They Think?” **ECML PKDD 2026**.
4. **JaeSeong Kim** and Suan Lee, “Pretraining Builds Cooperation, Not Components: Tracking Inter-Layer Coordination via Lie Brackets,” **ACL Rolling Review**, March 2026 (Findings).

JOURNAL ARTICLES (KCI)

1. **JaeSeong Kim** and Suan Lee, “Efficiently Lightweight Korean Language Model with Post-layer Pruning and Multi-stage Fine-tuning,” 2025.

2. **JaeSeong Kim**, SuYun Ko, JinGeun Jeong, and Suan Lee, “LLM as a Search: 사용자 의도 분석을 통한 지능형 법률 문서 검색,” 2025.
3. **JaeSeong Kim** and Suan Lee, “Improving Text-to-SQL Model Performance via Step-by-Step and Self-Debug Approaches,” 2025.

INTERNATIONAL CONFERENCES

1. **JaeSeong Kim** and Suan Lee, “How Language Directions Align with Token Geometry in Multilingual LLMs,” ACL Student Research Workshop (SRW), 2026.
2. **JaeSeong Kim** and Suan Lee, “Linear Separability and Feature Importance Analysis in Large Language Model Representation Spaces,” IEEE BigComp, 2026. **Best Paper Award Runner-Up.**
3. SangHyeon Gil, **JaeSeong Kim**, and Suan Lee, “Analyzing LLM Safety Circumvention via Multilingual Euphemistic Coating,” IEEE BigComp, 2026.

DOMESTIC CONFERENCES

1. SangHyeon Gil, **JaeSeong Kim**, and Suan Lee, “한국어 완곡 코팅을 통한 LLM 안전성 우회 분석,” IEIE, 2025. **Encouragement Award.**
2. **JaeSeong Kim** and Suan Lee, “Order Within LLMs: Analyzing Task-specific Representation Organization in Latent Spaces,” KCC, 2025.
3. **JaeSeong Kim** and Suan Lee, “KO-SmallThinker: Reasoning 기반 소형 언어 모델을 활용한 초거대 언어 모델의 성능 한계 극복,” KCC, 2025.
4. **JaeSeong Kim** and Suan Lee, “Minimal Tuning, Maximum Gains: 선택적 레이어 파인튜닝 기반 효율적 학습 전략,” KCC, 2025.
5. **JaeSeong Kim** and Suan Lee, “K-GovExam: 한국 공무원 시험 기반 LLM 평가용 데이터셋 구축 및 추론 언어 모델 분석,” KCC, 2025. **Grand Prize.**
6. **JaeSeong Kim** and Suan Lee, “Depth-Up Scaling 을 활용한 언어 모델 증강과 소규모 데이터 학습 성능 비교,” KISM, 2024.
7. **JaeSeong Kim** and Suan Lee, “Improving Text-to-SQL Model Performance via Step-by-Step Learning,” KISM, 2024.
8. **JaeSeong Kim**, KangJun Kim, and Suan Lee, “법률 QA 데이터셋을 이용한 거대 언어 모델 학습,” KISM, 2024.
9. **JaeSeong Kim** and Suan Lee, “파인튜닝, RAG, 프롬프트 기반 언어 모델의 성능 비교 분석,” KISM, 2024.
10. **JaeSeong Kim** and Suan Lee, “서로 다른 언어 모델의 상징적 지식 증류를 이용한 경량화된 감정 분석 모델,” KSC, 2023.

UNDER REVIEW

1. **JaeSeong Kim** and Suan Lee, “Semantic Directions Emerge from Sparse Token Mixture,” 2026. *(Under Review)*
2. **JaeSeong Kim** and Suan Lee, “Latent Sentinel: Real-Time Jailbreak Detection with Layer-wise Probes,” 2026. *(Under Review)*
3. **JaeSeong Kim** and Suan Lee, “A Hybrid Approach Combining Recurrent Structure and Attention Mechanism for Early Prediction of Real-time Streaming Text,” 2026. *(Under Review)*
4. **JaeSeong Kim** and Suan Lee, “Module Attack: Exploiting Module Swapping to Compromise LLM Alignments,” 2026. *(Under Review)*

Research Projects

한국연구재단 우수연구

한국연구재단 (NRF)

- 선형 LM Head 구조 기반 은닉 표현 섭동의 선형 프로젝션을 이용한 학습 없는 LLM 워터마킹 및 탐지

정부지원 R&D (TIPA / 중소벤처기업부)

금융권 Text-to-SQL 솔루션

(주) 범익

Oct. 2024 – Present

- 생성형 LLM 기반 Text-to-SQL 솔루션 (3 개년 과제)

공무원 수험생 AI 학습 플랫폼

(주) 지티어트랙션

Aug. 2024 – Jul. 2025

- LLM 기반 과목별 문제 생성, 오답 분석, 취약점 진단 학습 지원 플랫폼 (2 개년 과제)

“PitchScore” 멀티모달 IR 피칭 평가

(주) 피치스파트너스

Jul. 2025 – Present

- 멀티모달 AI 기반 IR 피칭 평가 및 피드백 시스템

산업체 연구 영역

지휘 결심 지원 및 가상 시뮬레이션

(주) 네비웍스

Oct. 2025 – Mar. 2026

- 군 지휘 결심 지원을 위한 온톨로지 및 에이전트 개발

Honors & Awards

- 2025 총장상, 세명대학교
- 2026 **Best Paper Award Runner-Up**, IEEE BigComp 2026
- 2025 **Grand Prize (Best Award)**, KCC 2025 — K-GovExam
- 2025 **Encouragement Award**, IEIE 2025 — LLM Safety via Euphemistic Coating
- 2025 **Special Contribution Award**, KCC 2025 — KIISE